

Advanced (Habanero)

Q1. What is the remainder when 24 is divided by 5?

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 0

Q2. Which number is divisible by 3 and 9?

- (A) 12
- (B) 15
- (C) 18
- (D) 20
- (E) 21

Q3. If $a = 12$ and $b = 3$, what is the remainder of $a \div b$?

- (A) 0
- (B) 1
- (C) 2
- (D) 3
- (E) 4

Q4. What is $48 \div 6$ with remainder?

- (A) 8 R 0
- (B) 7 R 6
- (C) 6 R 12
- (D) 8 R 6
- (E) 6 R 0

Q5. Which of the following is NOT a multiple of 10?

- (A) 20
- (B) 30
- (C) 31
- (D) 40
- (E) 50

Q6. If 18 is divided by 3, what is the quotient and remainder?

- (A) 6 R 0
- (B) 5 R 3
- (C) 4 R 6
- (D) 6 R 3
- (E) 3 R 3

Q7. Is 27 divisible by 9?

- (A) Yes
- (B) No
- (C) Maybe
- (D) Sometimes
- (E) Always

Q8. What is $50 \div 10$?

- (A) 4
- (B) 5
- (C) 6
- (D) 4 R 0
- (E) 5 R 0

Open Ended Questions (FRQ)

Q9. Perform the long division to find $72 \div 9$. Show all steps and give the remainder.

Q10. If $a = 25$ and $b = 4$, what is $a \div b$ with remainder? Express your answer using $a = qb + r$. Find q and r .

Chef's Tips

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- Remind students to check the divisibility rules before applying them
- Point out that long division should be performed step-by-step to avoid errors
- Suggest that students check their work by multiplying the quotient by the divisor and adding the remainder